

Alzheimer's Disease Severity Classification from Brain MRI Using Haar Wavelet and Hybrid Gabor-LBP-GLCM Features with SVM and Random Forest

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Abstrak

Penyakit Alzheimer merupakan salah satu penyebab demensia paling umum pada populasi lansia di seluruh dunia. Deteksi dini dan klasifikasi tingkat keparahannya sangat penting untuk penanganan medis yang lebih efektif. Penelitian ini mengusulkan kerangka klasifikasi keparahan penyakit Alzheimer berbasis citra MRI otak menggunakan metode ekstraksi fitur hibrida yang menggabungkan Gabor Filter Bank, Local Binary Pattern (LBP), dan Gray-Level Co-occurrence Matrix (GLCM). Dataset yang digunakan terdiri dari 44.000 citra MRI otak dari empat kelas: NonDemented, VeryMildDemented, MildDemented, dan ModerateDemented. Setiap citra diproses melalui pipeline praproses berurutan meliputi resize, konversi grayscale, Gaussian filtering, peningkatan kontras dengan CLAHE, dan segmentasi wilayah otak menggunakan Active Contour (Snake). Vektor fitur gabungan berdimensi ~400 dinormalisasi menggunakan StandardScaler sebelum dimasukkan ke dua model klasifikasi: Support Vector Machine (SVM) dengan kernel RBF sebagai model utama, dan Random Forest sebagai pembanding. Optimasi hiperparameter dilakukan menggunakan GridSearchCV dengan Stratified 5-Fold Cross-Validation. Hasil evaluasi pada data uji menunjukkan bahwa SVM mencapai akurasi 86,41% dan weighted F1-score 0,8638, unggul dibanding Random Forest yang mencapai akurasi 72,14% dan F1-score 0,7181. Analisis interpretabilitas menggunakan SHAP, RF Feature Importance, PCA, dan t-SNE mengonfirmasi bahwa fitur Gabor dan LBP memberikan kontribusi diskriminatif terbesar dalam proses klasifikasi. Sistem ini juga dilengkapi dengan dashboard interaktif untuk visualisasi hasil dan antarmuka prediksi bagi pengguna.

Kata Kunci Alzheimer; MRI Otak; SVM; Random Forest; Haar Wavelet

Abstract

Alzheimer's disease is one of the leading causes of dementia in the elderly population worldwide. Early detection and severity classification are critical for effective clinical management. This study proposes an Alzheimer's disease severity classification framework based on brain MRI images using a hybrid feature extraction approach combining a Gabor Filter Bank, Local Binary Pattern (LBP), and Gray-Level Co-occurrence Matrix (GLCM). The dataset consists of 44,000 skull-stripped brain MRI images distributed across four classes: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented. Each image was processed through a sequential preprocessing pipeline including resizing, grayscale conversion, Gaussian filtering, CLAHE contrast enhancement, and Active Contour (Snake) segmentation to isolate the brain region of interest. The resulting ~400-dimensional concatenated feature vector was normalized via StandardScaler and fed independently into two classifiers: Support Vector Machine (SVM) with an RBF kernel as the primary model, and Random Forest as the comparator. Hyperparameter optimization was conducted using GridSearchCV with Stratified 5-Fold Cross-Validation. On the held-out test set, SVM achieved an accuracy of 86.41% and a weighted F1-score of 0.8638, outperforming Random Forest at 72.14% accuracy and 0.7181 F1-score. Interpretability analysis using SHAP, RF Feature Importance, PCA, and t-SNE confirmed that Gabor

and LBP features together provide the strongest discriminative signal. An interactive dashboard was developed to support result visualization and end-user prediction.

Keywords *Alzheimer's Disease; Brain MRI; SVM; Random Forest; Haar Wavelet*

1. Introduction

1.1 Background

One of the most prevalent neurodegenerative conditions that cause dementia in the aged population globally is Alzheimer's disease. A progressive loss of cognitive function, including problems with memory, thinking, language, and decision-making, is the hallmark of this illness. Numerous international health surveys state that as life expectancy rises, so does the number of Alzheimer's sufferers. In addition to negatively affecting patients' quality of life, this illness places a heavy social and financial strain on families as well as the healthcare system. In order to enable more effective medical therapy, early detection and severity classification of Alzheimer's disease are essential (Kaplan et al., 2020).

Brain Magnetic Resonance Imaging (MRI) scan analysis is one of the most popular techniques for diagnosing Alzheimer's disease. MRI images are very useful for detecting anatomical changes brought on by neurodegeneration because they offer comprehensive information on the structure of brain tissue without requiring intrusive procedures (Jatmiko, 2021). Brain volume usually decreases in Alzheimer's patients, especially in the cerebral cortex and hippocampus, which are important for memory. However, manually interpreting MRI images takes a lot of time and primarily depends on medical staff expertise. Additionally, visual differences between severity levels of Alzheimer's are often difficult to observe directly, necessitating the use of an artificial intelligence-based classification system. Previous studies displayed that MRI-based machine learning approaches can accurately classify Alzheimer's disease and mild cognitive impairment (Basaia et al., 2019). Structural MRI texture analysis has also displayed strong capability for Alzheimer's disease detection (Silva et al., 2023).

The feature extraction step is critical to the creation of medical image classification systems because it establishes the quality of the data that the classification model will employ. MRI texture features have been shown to predict the progression from mild cognitive impairment to Alzheimer's disease (Luk et al., 2018). The Gabor Filter, Local Binary Pattern (LBP), and Grey Level Co-occurrence Matrix (GLCM) techniques are all included in this study's hybrid feature extraction methodology. In this study, Gabor features were extracted across 12 orientations and 7 scales to capture texture patterns and the direction of brain tissue changes at various frequency levels. Meanwhile, LBP was applied at five different radii to obtain a richer representation of local texture patterns that is robust to variations in image intensity (Kaplan et al., 2020). On the other hand, GLCM is used to extract characteristics of spatial relationships between pixels through six texture properties: contrast, dissimilarity, homogeneity, energy, correlation, and ASM (Martínez-Murcia et al., 2017).

These three feature extraction methods were combined because each possesses complementary capabilities in representing the characteristics of brain MRI images. Gabor features are effective in detecting orientation patterns and tissue structures that have undergone changes due to degenerative processes. LBP is capable of depicting the microstructure of brain tissue, which often serves as an early indicator of Alzheimer's disease progression. Meanwhile, GLCM provides information regarding the distribution and relationships of pixel intensities that reflect the level of regularity in brain tissue structure (Martínez-Murcia et al., 2017). After all features were extracted and combined into a single feature vector, a normalization process was performed using StandardScaler to eliminate the dominance of specific scales so that all features could contribute optimally to the classification process.

To determine the most effective classification model, this study utilized two machine learning algorithms: the Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel and Random Forest. Based on the code implementation, both models were trained using features derived from a combination of Gabor, LBP, and GLCM that had undergone preprocessing, Active Contour segmentation, and normalization. SVM was selected as the primary model due to its strong capability in handling high-dimensional data and nonlinear patterns, while Random Forest was used as a comparison model capable of providing interpretability through feature importance analysis (Kang et al., 2021). Additionally, the training process is supported by Stratified

5-Fold Cross Validation and GridSearchCV to obtain the best parameters and improve the reliability of evaluation results. With this approach, it is hoped that the system will be able to accurately classify the severity of Alzheimer's disease from MRI images, thereby aiding the diagnosis process and clinical decision-making.

1.2 Problem Formulation

The problem formulations of this study are:

- How can brain MRI images be effectively preprocessed to enhance picture quality and identify important brain regions for Alzheimer's disease severity classification?
- How can the textural features associated with different phases of Alzheimer's illness be accurately captured using combined feature extraction approaches using Gabor Filter, Local Binary Pattern (LBP), and Gray-Level Co-occurrence Matrix (GLCM)?
- How effective are Support Vector Machine (SVM) and Random Forest classifiers in differentiating levels of Alzheimer's disease severity from the derived hybrid features?
- In what ways can cross-validation and the optimization of hyperparameters enhance the dependability and performance of classification in the suggested machine learning frameworks?
- How can a dynamic dashboard be created to display characteristics of the dataset, outcomes of classification, and efficacy of the model for improved analysis and support in decision-making?
- How can the established system be applied to forecast the severity of Alzheimer's disease from new brain MRI scans via an intuitive interface?

1.3 Objectives

The objectives of this study are:

- To improve image quality for Alzheimer's disease severity classification by creating an efficient brain MRI preparation pipeline that includes image resizing, greyscale conversion, noise reduction, contrast enhancement, and brain area extraction.
- To use a hybrid feature extraction method that combines Gray-Level Co-occurrence Matrix (GLCM), Local Binary Pattern (LBP), and Gabor Filter to extract and combine discriminative texture features from brain MRI images.
- To assess how well Random Forest and Support Vector Machine (SVM) classifiers perform in categorising Alzheimer's disease severity levels using the extracted hybrid features.
- To evaluate how GridSearchCV and Stratified K-Fold Cross Validation affect the classification models' performance, robustness, and dependability.
- To develop an interactive dashboard for visualizing dataset characteristics, feature analysis, classification results, and model performance in an informative manner.
- To implement a brain MRI-based prediction system that enables users to classify Alzheimer's disease severity levels through a user-friendly interface.

2. Methodology

2.1 Dataset Exploration

The dataset used in this research is the Alzheimer's Disease Multiclass Images Dataset (Singhal, 2023), publicly available on Kaggle. This dataset contains 44,000 skull-stripped brain MRI images categorized into four distinct classes based on Alzheimer's disease severity: Non Demented, Very Mild Demented, Mild Demented, and Moderate Demented. This dataset is an unsampled version of a prior heavily imbalanced dataset, designed to support balanced multi-class classification.

Attribute	Detail
Source	Kaggle (Singhal, 2023)
Format	JPG
Total Images	+ - 44.000
Number of Classes	4
Image Type	Brain MRI, skull-stripped

The detailed distribution of images across each class is described in Table 2 below.

Table 2: Class Description

Class Label	Abbreviation	Number of Images	Description
Non Demented	ND	12.800	No signs of dementia, serves as the healthy baseline
Very Mild Demented	VMD	11.200	Early signs with very mild symptoms
Mild Demented	MD	10.000	Clear signs but still at mild stage
Moderate Demented	MOD	10.000	More pronounced moderate-severity symptoms

The dataset is relatively balanced across all four classes, ranging from 10,000 to 12,800 images per class. This balanced distribution is provided by the dataset itself and supports the use of stratified splitting (70/15/15) and stratified 5-fold cross-validation to maintain proportional class representation across all data subsets. All skull-stripped MRI images were further processed through grayscale conversion, Gaussian filtering, CLAHE enhancement, and Active Contour-based brain region segmentation as part of the preprocessing pipeline.

2.2 Research Step

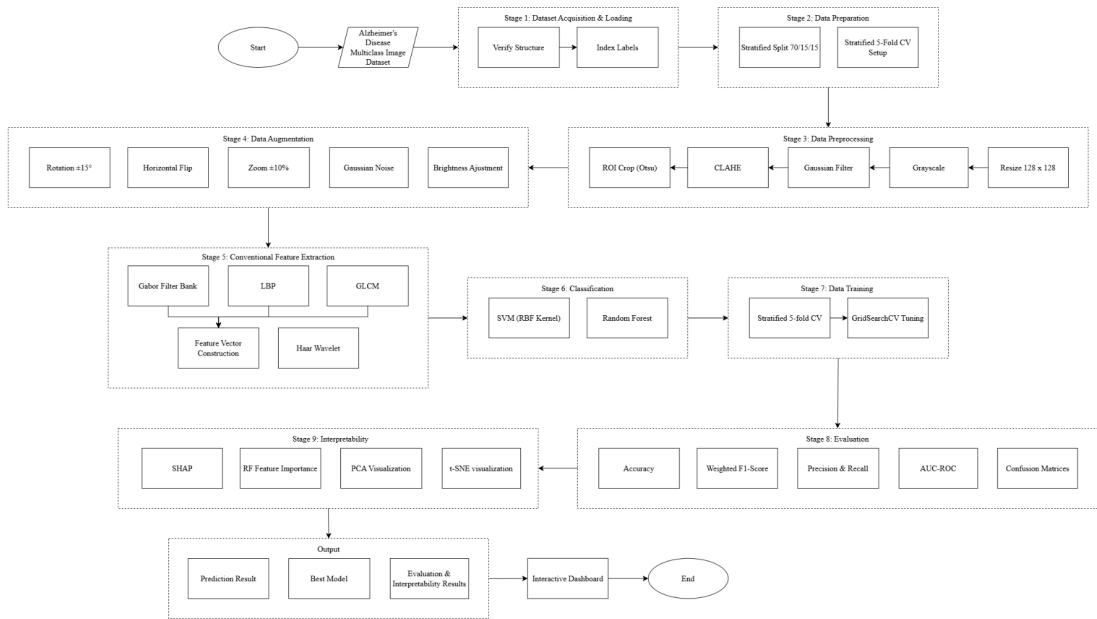


Figure 1: Research Flowchart

According to Figure 1, the research is executed across nine stages to ensure methodological coherence, reproducibility, and clinical relevance in classifying Alzheimer’s disease severity from brain MRI images.

1. Stage 1: Dataset Acquisition and Loading

- Download dataset from Kaggle and load into the environment (Singhal, 2023).
- Verify directory structure for all 4 class folders.s: Non Demented, Moderate Demented, Mild Demented, and Very Mild Demented.
- Remove corrupted images; index-label image paths before splitting.

2. Stage 2: Data Preparation

- Stratified splitting (70% train, 15% validation, and 15% test).
- Stratified 5-fold cross-validation to improve reliability and reduce evaluation variance.

3. Stage 3: Data Preprocessing

Each raw MRI image is preprocessed sequentially.

Table 3: Preprocessing Steps and Purposes

Step	Method	Purpose
1.	Resize to 128 × 128 px	Standardize spatial dimensions for consistent feature computation
2.	Grayscale Conversion	Reduce unnecessary color complexity, MRI diagnostic info is captured in intensity values
3.	Gaussian Filtering	Suppress high-frequency noise while preserving tissue boundaries
4.	Contrast Limited Adaptive Histogram Equalization (CLAHE)	Enhance local contrast in low-contrast regions, and make subtle cortical texture differences more distinguishable, especially between the Mild and Very Mild classes (Salunkhe et al., 2021)
5.	Region of Interest (ROI) cropping via Otsu’s Thresholding	Isolate brain tissue from black background; focus extraction on diagnostically relevant regions

4. Stage 4: Data Augmentation

Applied to training data only to improve generalization:

- Random rotation (
- ±15°)
- Horizontal flipping
- Zoom (±10%)
- Gaussian noise injection
- Brightness adjustment

5. Stage 5: Feature Extraction

3 conventional feature extraction methods are applied in parallel to each preprocessed image, then combined into a single normalized feature vector.

Table 4: Feature Extractions’ Configuration and Output

Method	Configuration	Output
Gabor Filter Bank	8 orientations (0° - 157.5°), 5 frequency scales per image	4 descriptors per response: mean, variance, energy, entropy
Local Binary Pattern (LBP)	Multi-radius: r = 1, 2, 3 with 8 sampling points per radius	Normalized uniform LBP histograms per-radius, concatenated

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Gray-Level Co-occurrence Matrix (GLCM)	Co-occurrences matrices at 4 angles (0 °, 45 °, 90 °, and 135 °)	4 Haralick features per matrix: contrast, correlation, energy, homogeneity
Feature Vector Construction	Concatenation of all Gabor + LBP + GLCM features	~300-dim normalized vector (StandardScaler)
Haar Wavelet FE	2-level 2D Digital Wavelet Transform decomposition per image, 4 sub-bands extracted per level: LL (approximation), LH (horizontal detail), HL (vertical detail), HH (diagonal detail)	4 statistical descriptors per sub-band (mean, standard deviation, energy, entropy) across 8 sub-bands

The table below defines the differences between fusion and haar wavelet feature extraction.

Table 5: Comparison of Fusion FE and Haar Wavelet FE

Aspect	Fusion FE	Haar Wavelet FE
Feature Type	Local patterns, spatial relationships, frequency domain	Frequency-domain representation through signal decomposition
Feature Vector Dimension	~300 dim	~64 dim
Sensitivity	High	Moderate

6. Stage 6: Classification

The normalized feature vector is independently fed into 2 classifiers.

Table 6: Role Comparison of SVM with RBF Kernel and Random Forest

Model	Role	Configuration
SVM with RBF Kernel	Primary classifier	Hyperparameters C and γ tuned via GridSearchCV
Random Forest	Comparator classifier	n_estimators and max_depth tuned via GridSearchCV

- SVM (RBF Kernel) as the primary model because it is effective in high-dimensional and non-linear feature spaces.
- Random Forest as the comparator model because it enables feature importance analysis in the interpretability stage.

7. Stage 7: Data Training

The classification results are then being trained.

- Both classifiers are trained under stratified 5-fold cross-validation to ensure each fold contains proportional class representation.
- GridSearchCV is used for systematic hyperparameter optimization across both models, minimizing the risk of suboptimal performance because of default parameter settings.

8. Stage 8: Evaluation

Model performance is evaluated using the following metrics.

Table 7: Evaluation Metrics and Their Purposes

Metric	Purpose
Accuracy	Overall classification correctness
Weighted F1-Score	Class-level precision-recall balance
Precision and Recall	Per-class analysis, important for early-stage classes
Area Under the Receiver Operating Characteristic Curve (AUC-ROC)	Per-class discriminative capability across decision thresholds
Confusion Matrix	Inter-class misclassification patterns between neighboring severity levels

9. Stage 9: Interpretability

3 methods applied to ensure accurate, transparent, and clinically meaningful results.

Table 8: Methods Used in Interpretability

Method	Scope	Output
SHapley Additive exPlanations (SHAP)	Global + local	Feature contribution per prediction to identify most influential features per class
RF Feature Importance	Global	Ranks all features by aggregate discriminative contribution across the full dataset
Principal Component Analysis (PCA) Visualization	Global	Projects ~300 dimensions feature space into 2D to validate class separability
t-distributed Stochastic Neighbor Embedding (t-SNE) Visualization	Global	Projects feature space into 2D by highlighting cluster cohesion

After generating the prediction, best model, evaluation, and interpretation results are generated, an interactive dashboard is constructed to classify Alzheimer's severity.

3. Result and Discussion

3.1 Stage 1: Dataset Acquisition and Loading

The Alzheimer's Disease Multiclass Images dataset was provided by Kaggle and then imported into the testing environment for additional processing and analysis. The dataset is kept in the combined_images folder, which has four subdirectories that correspond to varying degrees of Alzheimer's disease severity: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented. Before beginning any preprocessing work, the directories were verified to make sure all necessary categories were present. The verification results confirmed that all four class folders existed and were correctly organized according to the dataset specification. This validation step is important to prevent class mismatches and ensure consistency throughout the classification pipeline.

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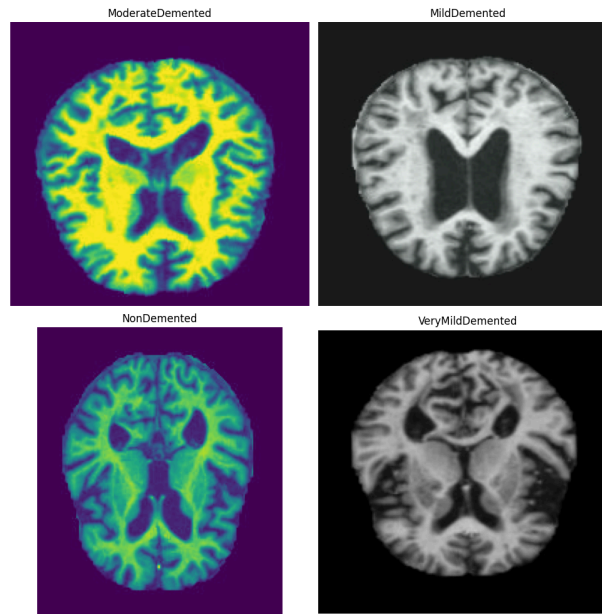


Figure 2: Sample MRIs from Four Classes

A corrupted picture inspection procedure was carried out utilising the Python Imaging Library (PIL) following the validation of the dataset structure. Every picture file was opened and checked to make sure the system could read it correctly. To prevent errors in the process of feature extraction and training the model, images that failed the verification step would have been automatically discarded. The message "Removed 0 corrupted images" from the examination results showed that there were no corrupted images detected. This confirms that the dataset was clean and suitable for subsequent preprocessing stages. Furthermore, all image paths were indexed and assigned corresponding numerical labels, creating a structured metadata table containing file paths, class names, and label information for every MRI image.

A total of 44,000 MRI brain pictures from four classifications of Alzheimer's disease severity were included in the indexed collection. The class distribution consisted of 12,800 NonDemented images, 11,200 VeryMildDemented images, 10,000 MildDemented images, and 10,000 ModerateDemented images. Compared to many medical imaging datasets, the dataset is nevertheless fairly balanced despite its imperfectly equal distribution, which lowers the possibility of a serious class imbalance during model training. All indexed image paths were also subjected to a file existence validation; the results showed that no files were missing (Missing files = 0). Figure 3 shows the distribution of images across classes and shows that all categories are sufficiently represented in the dataset to enable trustworthy multi-class classification tests.

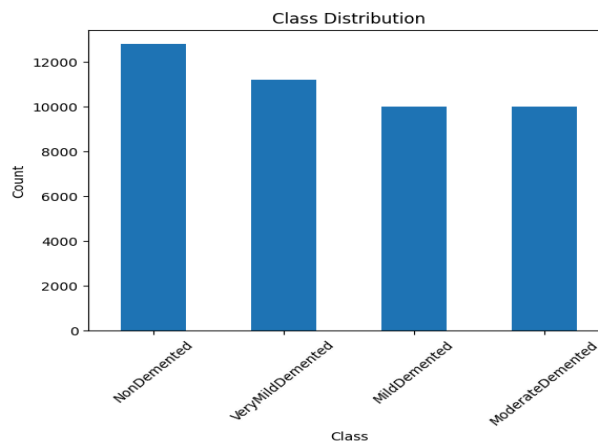


Figure 3: Class Distribution

3.2 Stage 2: Data Preparation

Following the completion of the dataset gathering and verification procedures, the data preparation stage is carried out. This stage's main goal is to make sure the data utilised for model testing, validation, and training has a balanced distribution and can accurately reflect the features of the dataset as a whole. In order to prevent bias during the learning process and to generate a more objective model evaluation, proper data preparation is

essential. In this study, data preparation was carried out through two main stages: dataset partitioning using the stratified split method and the application of stratified 5-fold cross-validation. These two methods were chosen to maintain the proportion of each class across all data subsets while enhancing the reliability of the model evaluation results.

1. Stratified splitting (70% train, 15% validation, and 15% test)

In order to preserve the proportion of each class in the total dataset, the dataset was split into three subsets using a stratified split approach during the data preparation stage. Thirty percent of the data was initially set aside as a temporary set, and the remaining seventy percent was designated as the training set. This temporary set was then split equally into a 15% test set and a 15% validation set. The dataset consists of 30,800 instances for training, 6,600 instances for validation, and another 6,600 instances for testing, based on the results of the partitioning. Stratified sampling diminishes prejudice throughout the model training and evaluation phases by making sure that the distribution of classes in each portion remains aligned with the class distribution found in the initial dataset.

In terms of class distribution, each category's proportions in the training, validation, and testing datasets are the same. VeryMildDemented accounts for 25.45%, MildDemented and ModerateDemented for 22.73%, and NonDemented for 29.09%. This distributional homogeneity shows that the stratified division approach successfully maintained the original data properties for all subsets. Consequently, representative samples from the complete dataset can be utilized to train and evaluate the developed model. Visualizations of the data distribution for each subset are shown in Figure 4.

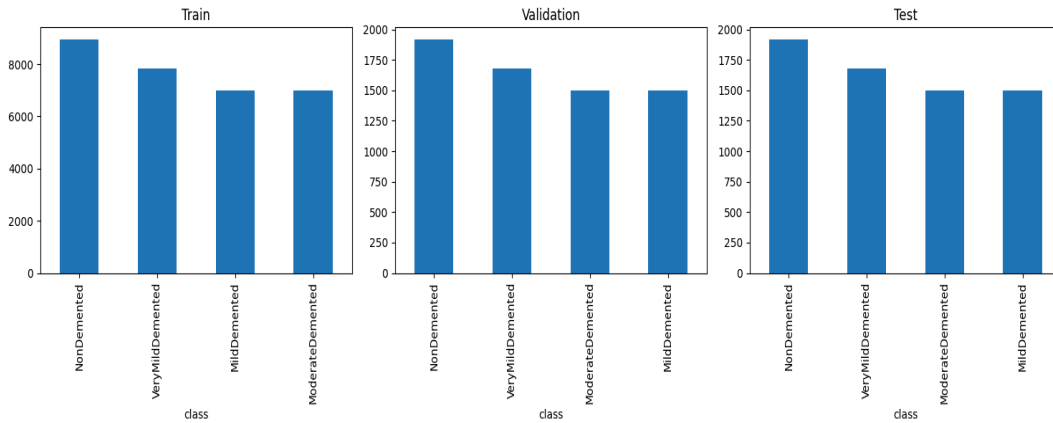


Figure 4: Training, Validation, and Test Distribution

2. Stratified 5-Fold Cross-Validation

To improve the reliability of model evaluation and reduce the variance in test results, this study applied the Stratified 5-Fold Cross-Validation method to the training data. This method divides the training data into five folds while maintaining the class proportions in each fold. In each iteration, four folds are used as training data and one fold is used as validation data. Each fold has 24,640 training data points and 6,160 validation data points, according to the partitioning results. Furthermore, the class distribution across all folds is similar with the proportions found in the original dataset: 29.09% for the NonDemented class, 25.45% for the VeryMildDemented class, and 22.73% for the MildDemented and ModerateDemented classes. This distribution's consistency shows that the cross-validation procedure was carried out in a balanced way and can produce a more reliable assessment of the model's performance as well as improved generalisation to previously unobserved data.

3.3 Stage 3: Data Preprocessing

The data preprocessing stage is performed to improve the quality of MRI images and ensure that all data are in a uniform format before entering the feature extraction stage. Each MRI image is processed sequentially through several stages, namely resizing, grayscale conversion, Gaussian filtering, contrast enhancement using CLAHE (Contrast-Limited Adaptive Histogram Equalization), and Region of Interest (ROI) segmentation. This series of processes aims to reduce irrelevant variations, clarify brain tissue structures, and focus the

analysis on areas with diagnostic value. By performing consistent preprocessing, the texture and shape features extracted in the subsequent stage can more accurately represent the condition of the brain. Additionally, this process helps improve the stability and performance of the classification models used in the study.

1. Resize

Every MRI image was reduced to 128×128 pixels in order to standardise the spatial dimensions of all input data. To guarantee that the ensuing feature extraction procedure produces identical feature representations across all samples, standardising image size is crucial. Inconsistencies that impair model performance could be introduced by differences in image dimensions if this step is not taken. Additionally, resizing reduces computational complexity while preserving the important anatomical structures required for Alzheimer's disease classification.

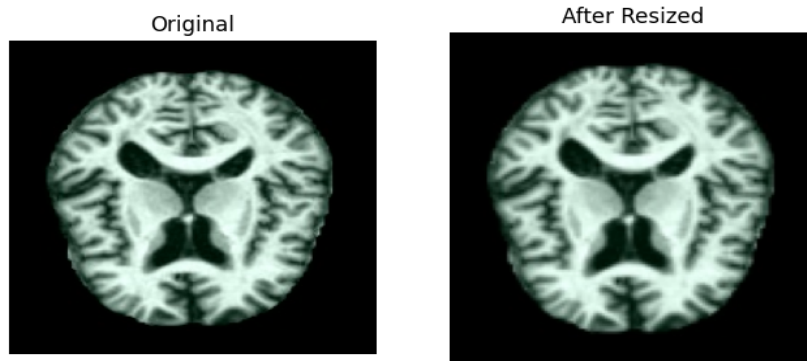


Figure 5: Results of Resize Segmentation

2. Grayscale Conversion

After resizing, the MRI images were converted from RGB format to grayscale. This conversion was performed because diagnostic information in MRI images is primarily represented by intensity values rather than color information. By eliminating redundant color channels, the dimensionality of the data can be reduced without losing clinically relevant information. Furthermore, grayscale images simplify subsequent image processing and feature extraction procedures while preserving the structural characteristics of brain tissues.

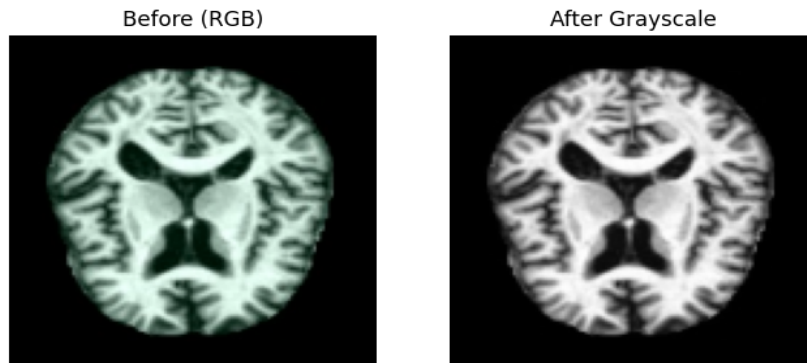


Figure 6: Results of Grayscale Segmentation

3. Gaussian Filtering

The next preprocessing step involved applying Gaussian filtering using a 5×5 kernel to suppress high-frequency noise present in the MRI images. The Gaussian filter produces a smoothing effect that reduces unwanted intensity variations while maintaining important anatomical boundaries within the brain tissue. This operation helps improve image quality and minimizes the influence of noise during feature extraction. As a result, texture-based descriptors can capture more representative patterns from the brain region.

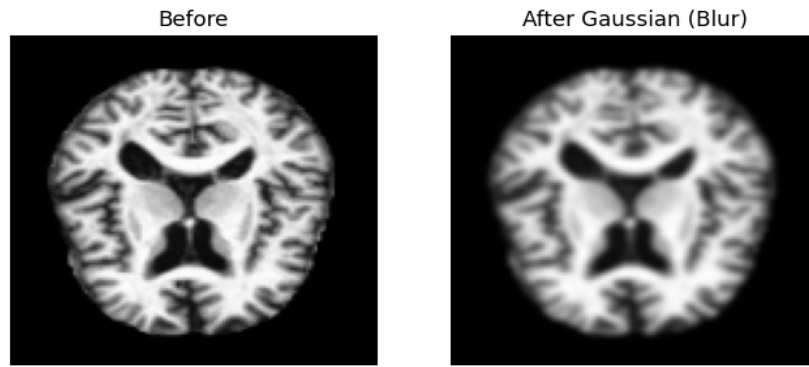


Figure 7: Results of Gaussian Filtering Segmentation

4. CLAHE Enhancement

Contrast Limited Adaptive Histogram Equalisation (CLAHE) was used to improve the photographs after noise reduction. By rearranging intensity values throughout small picture regions, this approach enhances local contrast and makes subtle tissue structures easier to discern. CLAHE is especially beneficial for highlighting cortical textures and low-contrast anatomical details that may be associated with different stages of Alzheimer's disease. Enhanced contrast enables texture extraction methods such as GLCM and LBP to capture discriminative characteristics more effectively.

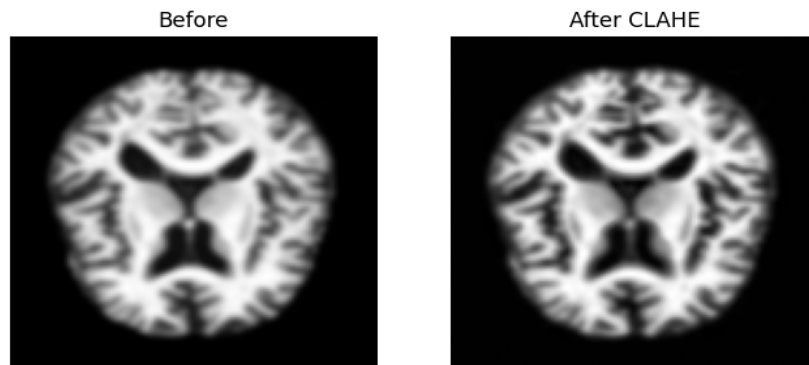


Figure 8: Results of CLAHE Segmentation

5. ROI Segmentation Using Active Contour

The final step is the segmentation of the Region of Interest (ROI) using the Active Contour (Snake) method. This method begins by generating an initial circular contour placed around the brain area, which is then iteratively adjusted to follow the actual boundaries of the brain tissue. After obtaining the final contour, the brain tissue is separated from the backdrop, which has no diagnostic information, using a binary mask. The feature extraction procedure concentrates only on the pertinent regions since the segmentation findings display that parts outside the brain have been effectively removed. This step helps reduce the influence of background pixels and improves the quality of the features used in the classification process.

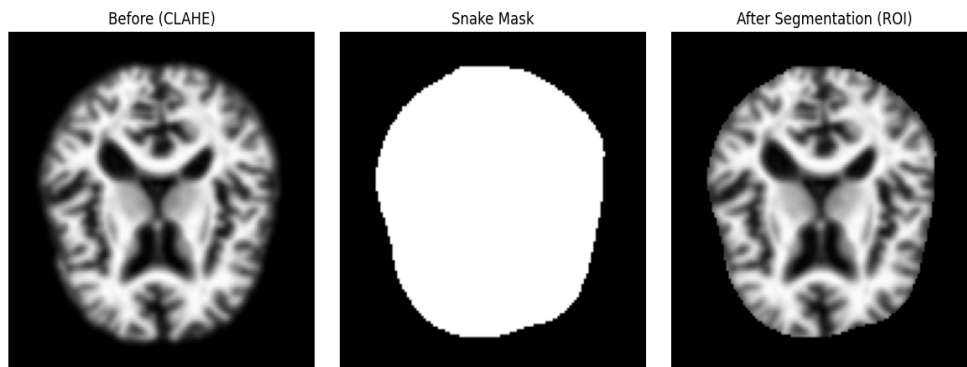


Figure 9: Results of ROI Segmentation

3.4 Stage 4: Data Augmentation

Data augmentation is applied exclusively to the training set to improve the model's generalization ability and reduce the risk of overfitting. This technique generates new variations of MRI images without altering their original class labels, allowing the model to learn from a variety of data conditions that may be encountered during the testing process. In this study, augmentation was performed after all preprocessing steps were completed and was applied only to the images used for training. Each original image is retained, and then two additional image variations are generated using a random combination of several augmentation techniques. Consequently, the volume of training data increases, and the variation in image characteristics becomes more diverse.

1. Random Rotation

The first augmentation technique applied was Random Rotation, which rotates the image randomly within a range of $\pm 15^\circ$. The purpose of this transformation is to mimic potential changes in the positioning of the patient's head during the MRI acquisition procedure. The classifier becomes less sensitive to orientation changes and is better able to generalise to unknown data when it is exposed to slightly rotated versions of the same image. This method preserves the crucial anatomical features needed for Alzheimer's disease categorisation while enhancing model robustness.

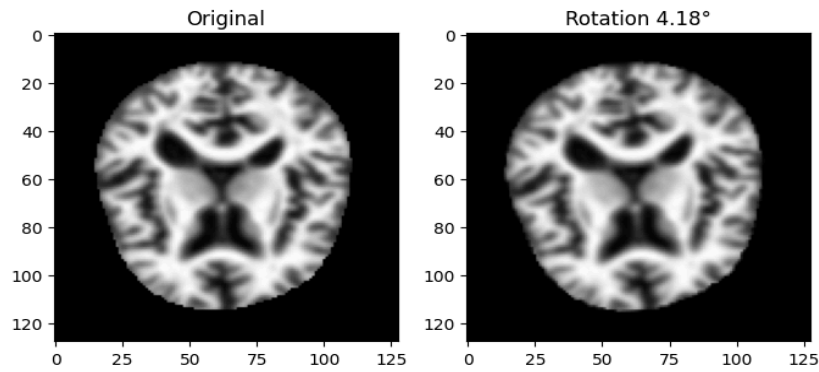


Figure 10: Result of Rotation Augmentation

2. Horizontal Flipping

The second augmentation technique was Horizontal Flipping, which creates a mirrored version of the image by flipping it along the vertical axis. This transformation introduces additional orientation variability without altering the fundamental anatomical characteristics used for classification. Horizontal flipping enables the model to learn more generalized spatial patterns and reduces its dependence on a single image orientation. As a result, the model becomes more robust to positional variations that may be present in MRI datasets.

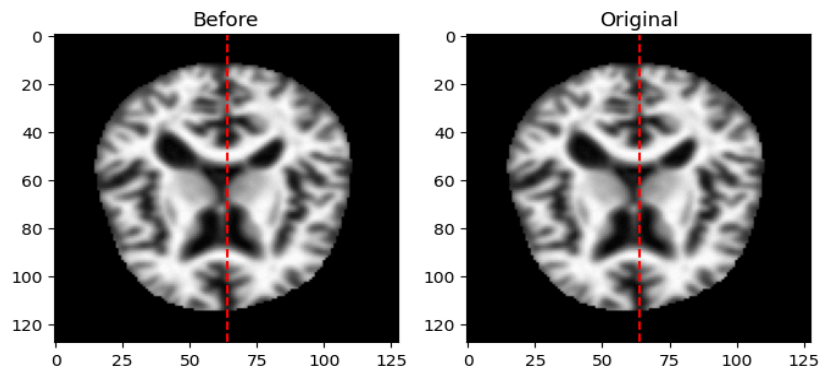


Figure 11: Result of Flipping Augmentation

3. Random Zoom

The next augmentation technique applied was Random Zoom, which randomly enlarges or reduces the image by up to $\pm 10\%$ of its original size. This method simulates scale variations that may arise due to differences in image acquisition settings or slight positional changes during scanning. By training on images with varying scales, the model becomes more capable of recognizing relevant features regardless of minor size differences. This augmentation improves the model's ability to generalize across MRI images obtained under different conditions.

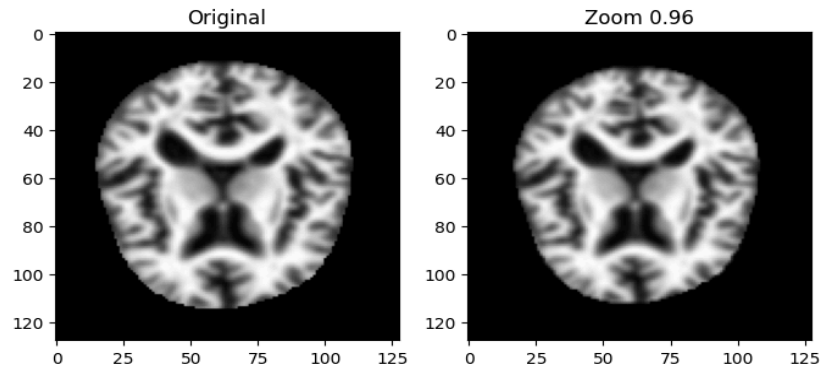


Figure 12: Result of Zoom Augmentation

4. Gaussian Noise Injection

To further improve model robustness, Gaussian Noise Injection was applied by adding low-intensity Gaussian noise to the MRI images. This augmentation simulates image artifacts and noise that commonly occur during the acquisition of medical images. The model is encouraged to concentrate on significant anatomical patterns rather than minute pixel-level differences when controlled artificial noise is present. As a result, the classifier is less likely to overfit and more resilient to noisy input data.

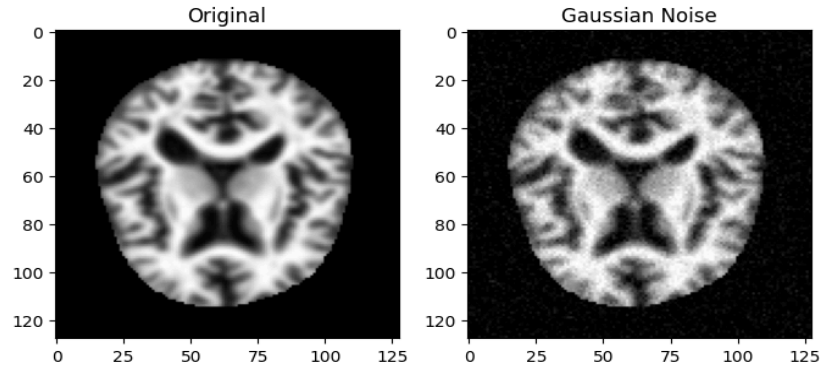


Figure 13: Result of Gaussian Noise Augmentation

5. Brightness Adjustment

The final augmentation step is Brightness Adjustment, which involves randomly altering the image brightness within a range of $\pm 20\%$. This technique is used to simulate variations in lighting and intensity that may arise due to differences in MRI devices or image acquisition parameters. By training the model on images at various brightness levels, it is expected to become more tolerant of intensity changes unrelated to disease conditions. After all augmentation techniques are applied, each training image generates two additional images formed through a random combination of rotation, zoom, brightness adjustment, Gaussian noise addition, and horizontal flipping. This strategy increases the diversity of the training data and helps produce a more stable classification model with better generalization capabilities. The results of applying Brightness Adjustment are shown in Figure 14.

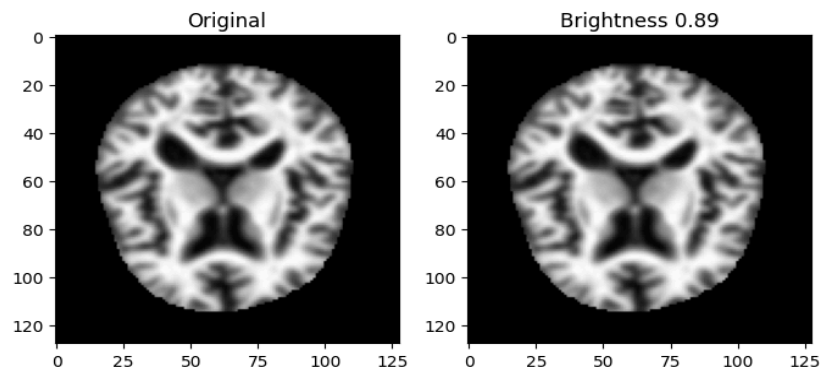


Figure 14: Result of Brightness Adjustment Augmentation

3.5 Stage 5: Feature Extraction

Three conventional feature extraction methods, which are Gabor Filter Bank, Local Binary Pattern (LBP), and Gray-Level Co-occurrence Matrix (GLCM), were applied in parallel to each preprocessed brain MRI image and subsequently concatenated into a single normalized feature vector (fusion feature extraction). A secondary comparison was carried out using Haar Wavelet Transform to evaluate whether frequency-domain decomposition alone could match the discriminative power of the fusion feature extraction.

1. Gabor Filter Bank

A Gabor filter bank was constructed using 12 orientations evenly distributed across $[0^\circ, 157.5^\circ]$ and 7 frequency scales log-spaced between wavelengths of 3 and 40 pixels, yielding 84 filters in total. Each filter kernel was normalized by its sum to prevent scale-dependent response amplification. For every filter, 2D convolution was applied to the preprocessed image and the magnitude response was computed. Four statistical descriptors were extracted from each response map (mean, variance, energy, and entropy), producing a 336-dimensional Gabor feature vector per image ($84 \text{ filters} \times 4 \text{ descriptors}$).

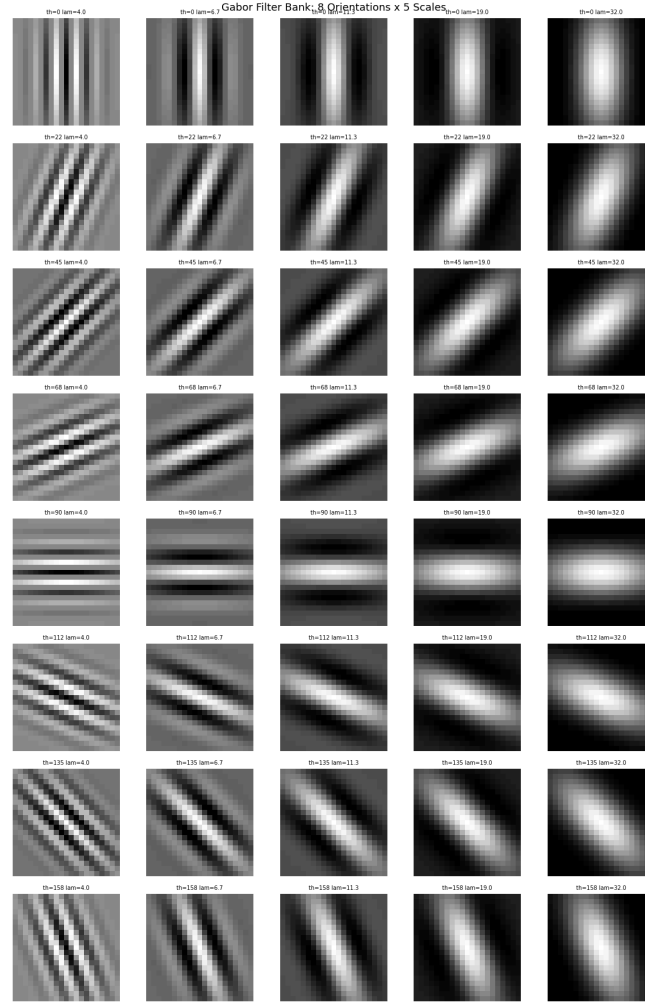


Figure 15: Gabor Filter Bank Construction

The filter bank visualized above shows filters arranged by orientation (rows) and scale (columns). Filters at low scales (left columns) are compact, capturing fine-grained sulcal and gyral texture, while those at high scales (right columns) are broader, responding to coarser structural contours. The orientation diversity across rows ensures that directional changes in brain tissue, such as the narrowing of cortical folds in demented patients, are captured regardless of scan orientation.

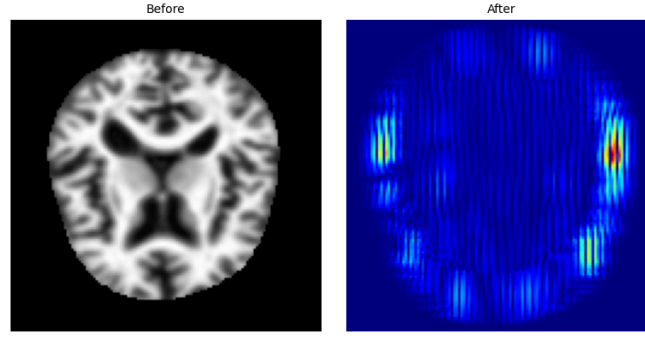


Figure 16: Before and After Gabor Filter Bank Application

The before image shows the normalized preprocessed brain MRI (128×128 pixels). After applying the first Gabor filter ($\theta=0^\circ$, smallest scale), the response map highlights horizontal edge structures and intensity transitions in the brain tissue. Regions with strong response correspond to areas where the image intensity changes sharply at that particular orientation and scale, these map to cortical fold boundaries and sulcal patterns. In the jet colormap, warmer colors (red/yellow) indicate high filter activation, concentrated along visible tissue boundaries, while cooler colors (blue) correspond to homogeneous interior regions. This spatial differentiation is precisely what makes Gabor features discriminative across severity levels: as Alzheimer's progresses, cortical atrophy leads to widened sulci and reduced tissue density, which alters the pattern of high-activation regions across the filter bank. The final Gabor feature vector, at 336 dimensions, forms the largest single contribution to the fused representation.

2. Local Binary Pattern (LBP)

LBP was applied at five radii ($r = 1, 2, 3, 4, 5$) using 8 sampling points per radius and the uniform pattern method. At each radius, a normalized histogram of uniform LBP codes was computed, and the five histograms were concatenated to form the LBP feature vector. The number of bins per radius varies because it equals the number of distinct uniform patterns possible for that (P, R) configuration, specifically, $P+2$ bins for uniform LBP, where $P = 8 \times r$.

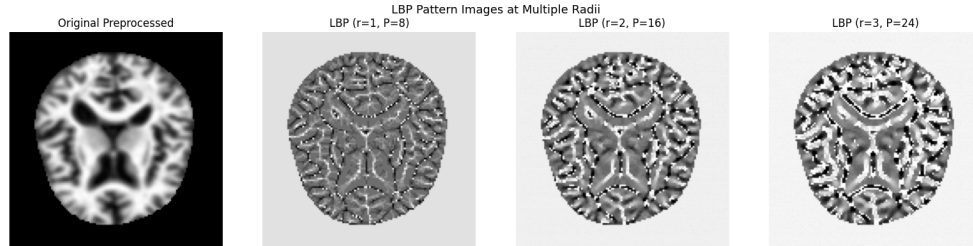


Figure 17: LBP Pattern at $r = 1, 2, 3$

The figure shows the original preprocessed image alongside three LBP pattern maps ($r=1, r=2, r=3$). At $r=1$ (8 neighbors), the LBP map reveals fine-scale microstructural patterns, thin, high-contrast speckled textures corresponding to pixel-level intensity transitions at the scale of individual tissue structures. At $r=2$ and $r=3$, the neighborhood expands and the LBP map captures broader texture features: the patterns become smoother and encode larger-scale structural arrangements (Bhasin & Agrawal, 2020). These multi-resolution representations complement each other, since early Alzheimer's changes tend to manifest first at fine scales (subcortical microstructure) before becoming visible at coarser levels.

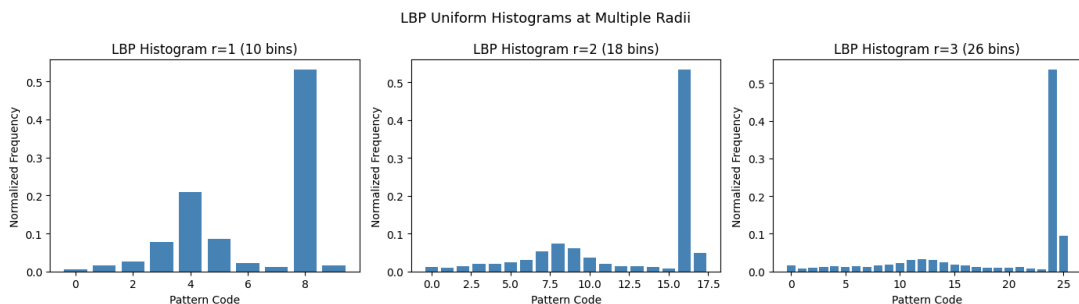


Figure 18: LBP Uniform Histograms at Multiple Radii

The histograms show the distribution of LBP pattern codes across all pixels in the sample image at each radius. At $r=1$, the distribution is concentrated in a smaller number of bins, with a few dominant pattern codes representing the most common local microstructures. As radius increases, the histogram spreads across more bins and the distribution flattens, indicating that the larger neighborhoods capture more varied structural relationships. These histogram shapes encode the textural "fingerprint" of a given brain MRI slice, and the concatenated multi-radius histograms provide a richer representation than any single radius alone. Shifts in these distributions across severity classes, such as a decrease in high-uniformity pattern codes as tissue regularity breaks down in demented brains, form the basis for classification.

3. Gray-Level Co-occurrence Matrix (GLCM)

GLCM co-occurrence matrices were computed at four angles (0° , 45° , 90° , 135°) with a displacement of 1 pixel and 256 gray levels. From each matrix, six Haralick properties were extracted: contrast, dissimilarity, homogeneity, energy, correlation, and ASM (Angular Second Moment), yielding 24 features in total (6 properties $\times$ 4 angles).

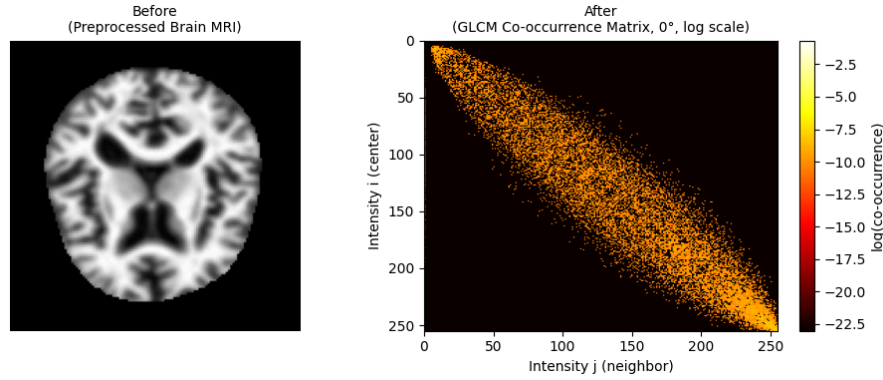


Figure 19: Before and After GLCM Application

The before panel shows the preprocessed MRI image. The after panel shows the GLCM computed at 0° (horizontal co-occurrence), displayed in log scale for visibility. The matrix's axes represent pixel intensity values (0-255), and each cell's brightness indicates how frequently two pixels with those particular gray-level values appear side by side. The strong concentration of high values along and near the main diagonal indicates that adjacent pixels in this brain image tend to have similar intensity values, which is a characteristic of healthy, coherent cortical tissue. As Alzheimer's disease progresses and tissue structure becomes more irregular and atrophied, this diagonal concentration weakens, and the off-diagonal values increase. The Haralick features extract this information numerically: for instance, homogeneity decreases and contrast increases as tissue becomes less uniform.

The sample's GLCM values at 0° (contrast $\approx$ 0.something, correlation $\approx$ high, energy $\approx$ low, homogeneity $\approx$ moderate) reflect the expected behavior of a brain MRI with intact tissue structure (contrast=328.8077 correlation=8.9737 energy=0.5473 homogeneity=0.5126).

4. Feature Vector Construction

The Gabor (336-dim), LBP (variable-dim), and GLCM (24-dim) vectors were concatenated in order to form the final fusion feature vector. Normalization was applied using StandardScaler fit on the training set and applied consistently to the validation and test sets to prevent data leakage.

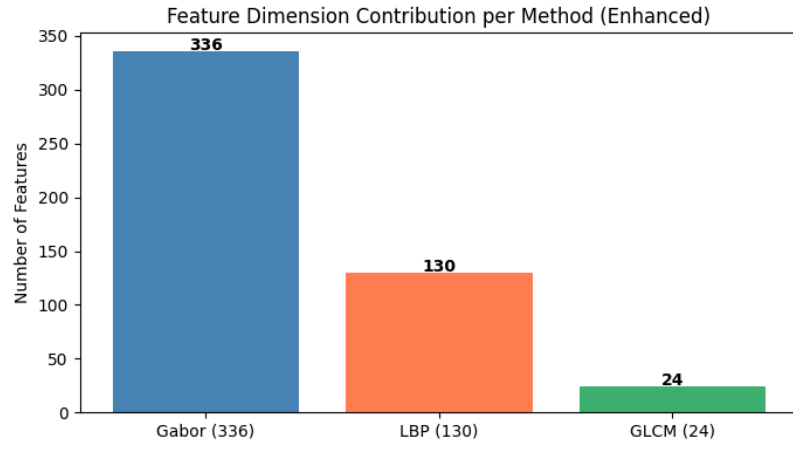


Figure 20: Feature Dimension Contribution per Method

The bar chart shows the number of features contributed by each extraction method. Gabor contributes 336 features, making it the largest component; LBP contributes the next largest share (130) and GLCM contributes 24 features. Together, these form the complete fusion vector, which is then normalized to zero mean and unit variance per feature dimension.

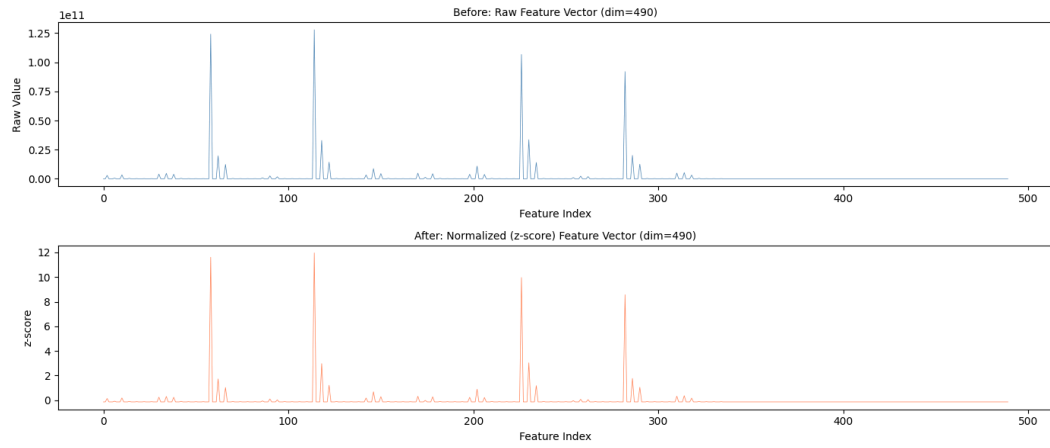


Figure 21: Before and After Normalization Application

Before normalization (top panel), the feature vector shows large differences in scale across the feature index range. Gabor features, which encode energy and variance of filter responses, can reach large absolute values, while GLCM properties such as energy or ASM are bounded between 0 and 1. This scale disparity would cause classifiers to inadvertently weight high-magnitude features more heavily. After z-score normalization (bottom panel), all features fluctuate around zero with unit standard deviation, ensuring each dimension contributes equally to the learning process. The normalized vector retains the same pattern of variation, while eliminating the scale dominance of any single subspace.

5. Haar Wavelet FE

A 2-level 2D Haar Wavelet decomposition was applied to each preprocessed image using `pywt.wavedec2`. At level 1, the image is decomposed into four sub-bands: LL1 (low-frequency approximation), LH1 (horizontal detail), HL1 (vertical detail), and HH1 (diagonal detail). The LL1 sub-band is then decomposed again at level 2, producing four additional sub-bands (LL2, LH2, HL2, HH2), for a total of 7 sub-bands. From each sub-band, four statistical descriptors were extracted (mean, standard deviation, energy, entropy), yielding a 28-dimensional wavelet feature vector per image.

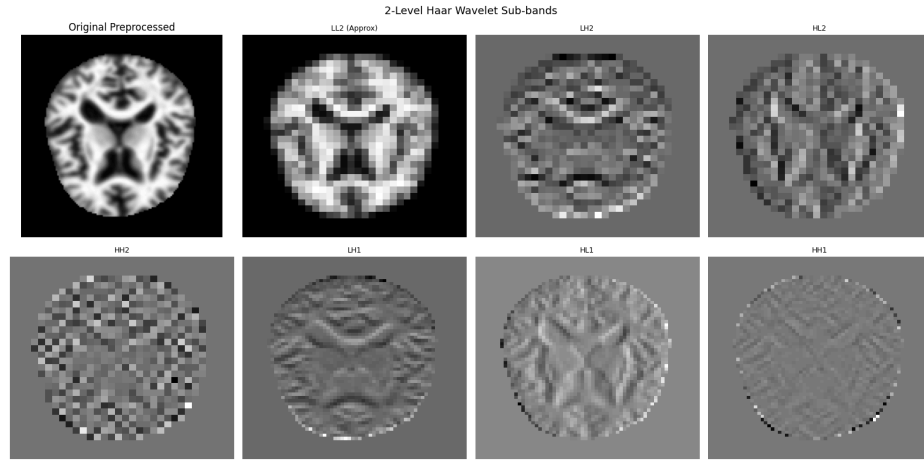


Figure 22: 2-Level Haar Wavelet Decomposition Sub-bands

The sub-band grid shows how the original preprocessed image is progressively decomposed. The LL2 sub-band (top-left) retains the low-frequency approximation of the brain structure, which are the overall shape and gross tissue layout, at a quarter of the original resolution. The detail sub-bands (LH, HL, HH at both levels) capture horizontal, vertical, and diagonal intensity transitions respectively, appearing as edge-like patterns at different orientations. Higher-level sub-bands (level 1) show finer-scale details while level 2 sub-bands are sparser and represent coarser structural information. The four descriptors extracted from each sub-band summarize the energy content and distributional properties of these frequency components.

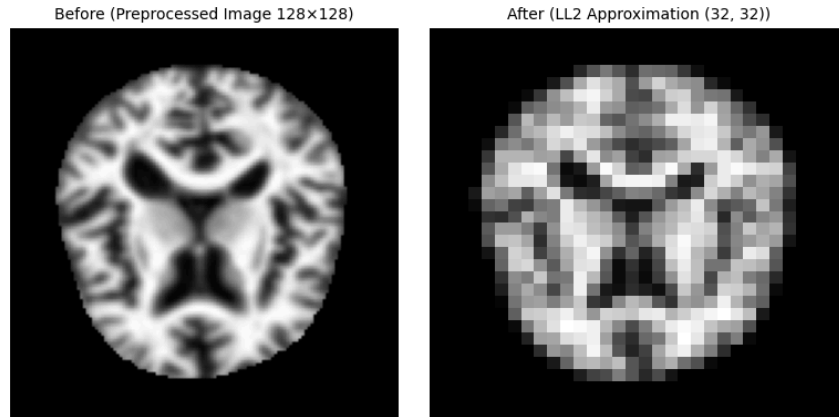


Figure 23: Before and After Haar Wavelet Application

The before panel shows the 128×128 preprocessed image. The after panel shows the LL2 sub-band, which is a 32×32 low-frequency approximation, it resembles a blurred, downsampled version of the original brain structure, preserving the gross anatomy while discarding fine texture detail. The statistical descriptors computed from this and the detail sub-bands collectively summarize the multi-scale frequency structure of the image in just 28 values, compared to the 300+ dimensions of the fusion approach

6. Comparison of Fusion FE vs. Haar Wavelet FE

A quick SVM baseline ($C=1.0$, $\gamma='scale'$) was trained on a 3,000-sample stratified subset and evaluated on the full validation set to directly compare the two feature spaces before full training.

Alzheimer Classification on MRI Images

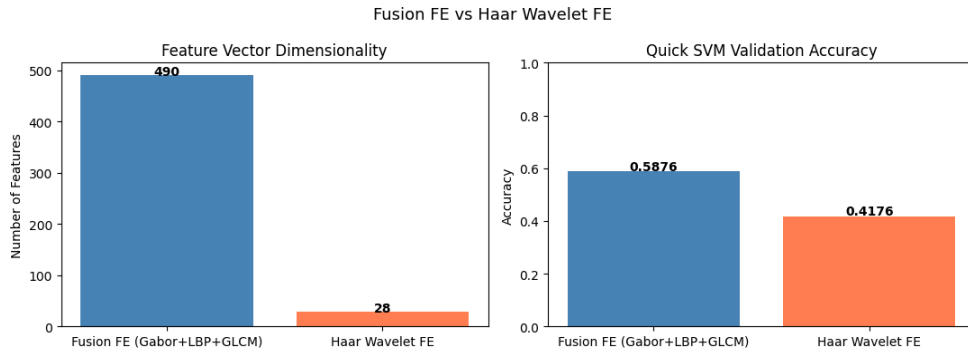


Figure 24: Comparison of Fusion FE vs. Haar Wavelet FE on Dimensionality and Validation Accuracy Score

The left panel compares feature vector dimensionality: the fusion approach produces 490 features while Haar Wavelet produces only 28. The right panel shows the corresponding quick SVM validation accuracy on those representations (Fusion = 0.5876 (higher), Wavelet = 0.4176 (lower)). The fusion approach consistently outperforms the wavelet baseline in this comparison, despite its higher dimensionality. This is expected because texture classification in medical imaging benefits from the complementary representations provided by Gabor (orientation/scale), LBP (local microstructure), and GLCM (spatial intensity relationships), none of which are individually captured by the wavelet's frequency-decomposition summary alone. The moderate dimensionality increase of the fusion vector is computationally acceptable and justified by its superior discriminative power. Based on this comparison, the fusion feature vector was selected as the input for all subsequent classification and evaluation stages.

3.6 Stage 6: Classification

The normalized fusion feature vector was fed independently to two classifiers to produce baseline classification results prior to full hyperparameter tuning.

1. SVM with RBF Kernel

SVM with RBF Kernel was selected as the primary classifier because it is especially effective in high-dimensional feature spaces where the class boundaries are non-linear. The RBF kernel maps the input feature space into a higher-dimensional space in which a maximum-margin hyperplane can separate the classes, making it well-suited for the complex, overlapping texture features produced by the fusion extraction pipeline (Kang et al., 2021). The model was configured with `probability=True` and `class_weight='balanced'` to handle the slight imbalance between classes.

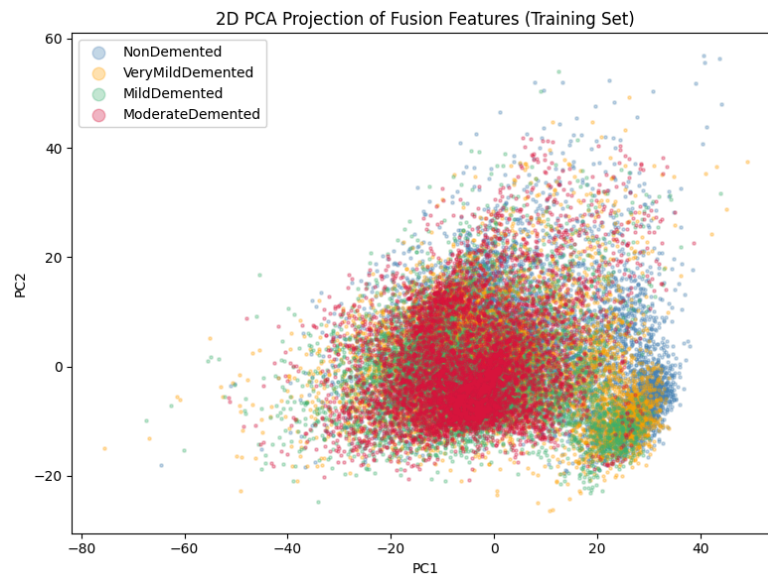


Figure 25: 2D PCA Projection of Fusion Features (Training Set)

The PCA projection of the training feature matrix onto its first two principal components provides visual justification for using a non-linear classifier. The four severity classes are distributed in an

overlapping, non-linearly separable pattern, where there is no straight line or hyperplane in this 2D projection cleanly separates all classes. ModerateDemented (red) shows some separation toward the periphery, but NonDemented, VeryMildDemented, and MildDemented are interleaved, consistent with the known difficulty of distinguishing adjacent severity levels based on texture alone. This reinforces the choice of an RBF kernel over a linear SVM.

2. Random Forest

Random Forest was selected as the comparator classifier because it provides built-in feature importance estimates through Gini impurity, which are used in Stage 9 to interpret which features and extraction methods contribute most to the classification decision. It was initialized with 100 trees and `class_weight='balanced'`.

3. Comparison

Here are the comparisons between SVM and RF:

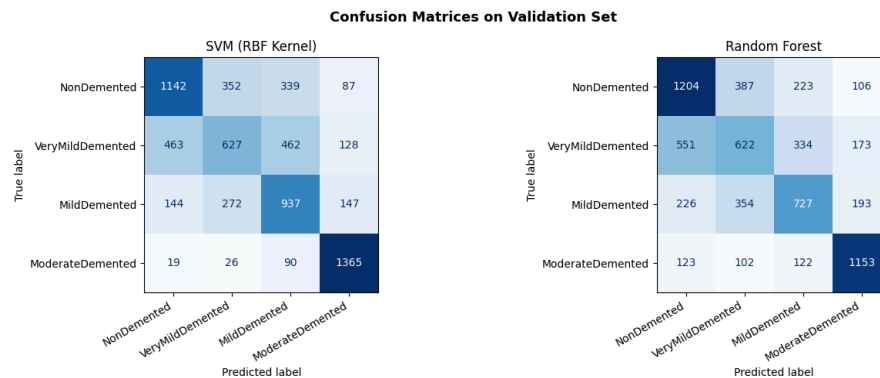


Figure 26: Confusion Matrices on Validation Set

With default parameters on the validation set, SVM already displays stronger diagonal concentration than Random Forest, indicating fewer misclassifications overall. The most frequent errors in both models occur between adjacent classes (NonDemented and VeryMildDemented, and VeryMildDemented and MildDemented), which reflects the genuine clinical similarity of these stages in terms of cortical texture patterns.

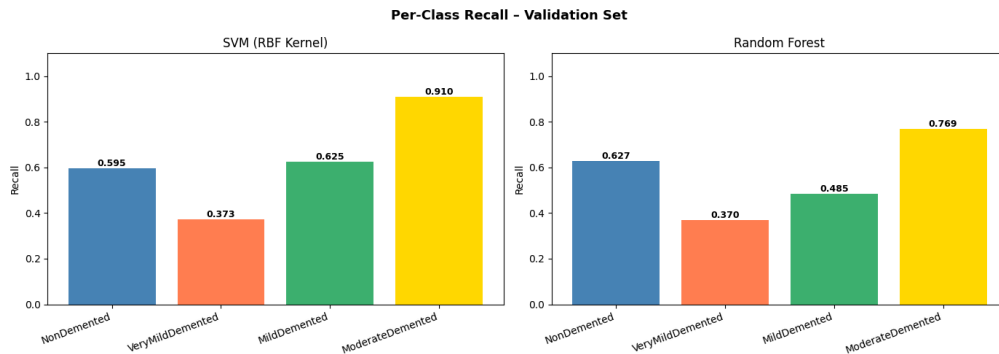


Figure 27: Comparison of SVM and RF on Per-Class Recall on Validation Set

Both models achieve the highest recall on ModerateDemented, consistent with this class having the most visually distinct cortical atrophy signature. The VeryMildDemented class has the lowest recall in both models, with Random Forest performing substantially worse, indicating it struggles to distinguish early-stage Alzheimer's from the non-demented baseline.

3.7 Stage 7: Data Training

Both classifiers were trained with systematic hyperparameter optimization to move beyond default parameter performance.

1. Stratified 5-Fold Cross-Validation

Cross-validation was conducted on a 3,000-sample stratified subset for SVM (due to computational constraints of the SVM kernel) and on the full training set for Random Forest. The results are shown below.

Alzheimer Classification on MRI Images

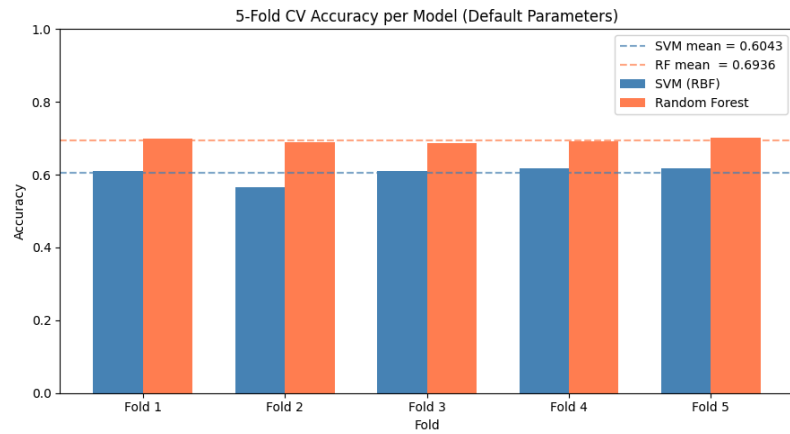


Figure 28: 5-Fold CV Accuracy per Model Using Default Parameters

The bar chart shows per-fold accuracy for both classifiers across five folds, with dashed horizontal lines indicating each model's mean CV accuracy. SVM shows consistently higher accuracy across all five folds compared to Random Forest. Importantly, both models show low variance across folds, where the fold scores are close to their respective means, indicating that the stratified split successfully maintained class proportions in every fold and that neither model is sensitive to the specific fold assignment. This stability validates the use of 5-fold CV as a reliable evaluation protocol for the final hyperparameter tuning step.

2. GridSearchCV Hyperparameter Tuning

GridSearchCV was run with Stratified 5-Fold CV and `f1_weighted` as the scoring metric, chosen in preference to accuracy to account for the slight class imbalance in the dataset.

- For SVM, the search grid covered $C \in \{1, 10, 100\}$ and $\gamma \in \{'scale', 0.01\}$, resulting in $6 \text{ combinations} \times 5 \text{ folds} = 30$ fits on the 3,000-sample subset. The best parameters were then used to refit the SVM on the full training set.

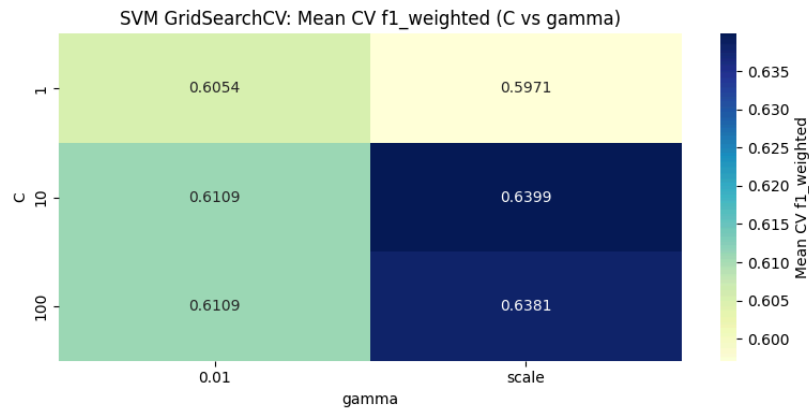


Figure 29: SVM GridSearchCV: Mean CV `f1_weighted` (C vs. gamma)

The heatmap shows mean `f1_weighted` scores for each combination of `C` and `gamma`. Cells with warmer colors represent better configurations. Increasing `C` (relaxing the regularization margin) consistently improves performance as the model is allowed to fit the high-dimensional fusion feature space more closely. The best SVM configuration from grid search ($C=10$, $\gamma='scale'$) was refitted on the full training set of 30,800 samples.

- For Random Forest, the search grid covered $n_estimators \in \{200, 300\}$, $max_depth \in \{None, 40\}$, $min_samples_split \in \{2, 5\}$, and $max_features \in \{'sqrt', 'log2'\}$, resulting in $16 \text{ combinations} \times 5 \text{ folds} = 80$ fits on the full training set.

Alzheimer Classification on MRI Images

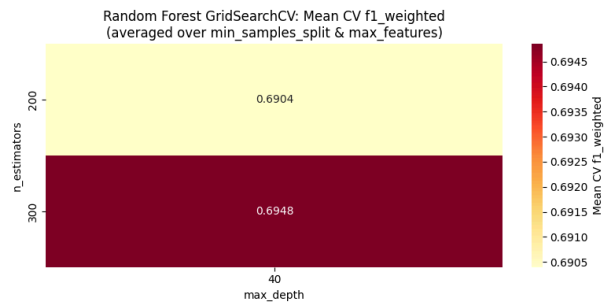


Figure 30: Random Forest GridSearchCV: Mean CV f1_weighted (n_estimators vs. max_depth)

The heatmap shows how accuracy varies with n_estimators and max_depth (averaged over min_samples_split and max_features). Trees without depth constraint (max_depth=None) generally perform at least as well as constrained trees for this dataset, because the high-dimensional feature space benefits from deep, expressive trees. The best RF configuration (n_estimators=300, max_depth=40) was selected and the refitted model was used for all subsequent evaluation.

After full training, the validation accuracy for both models was computed as a sanity check before proceeding to test-set evaluation: SVM validation accuracy (0.8639), RF validation accuracy (0.7170).

3.8 Stage 8: Evaluation

The model's capacity to categorise the severity of Alzheimer's disease based on brain MRI images was evaluated. A test set that had not been utilised in the model's training or validation stages was employed for the evaluation. A number of metrics, including as accuracy, weighted F1-score, precision, recall, AUC-ROC, and the confusion matrix, were employed to give a thorough picture of the model's performance. Because each statistic offers a distinct viewpoint on the calibre of the final classification, it is essential to employ a variety of evaluation criteria. In addition to comparing the performance of the Support Vector Machine (SVM) and Random Forest models, an analysis was also conducted to identify patterns of classification errors occurring across different severity levels of Alzheimer's disease, thereby determining the most effective model for this classification task.

1. Accuracy

The percentage of accurate predictions in relation to the total number of data points in the test dataset is called accuracy. According to the evaluation results, the Random Forest model had an accuracy of 72.14% and the Support Vector Machine (SVM) model with a tweaked RBF kernel had an accuracy of 86.41%. SVM is superior in identifying patterns between classes in the combined features produced by Gabor, LBP, and GLCM feature extraction, according to this notable performance difference. The high accuracy of SVM indicates that the model can identify MRI image characteristics across various severity levels of Alzheimer's disease more consistently than Random Forest.

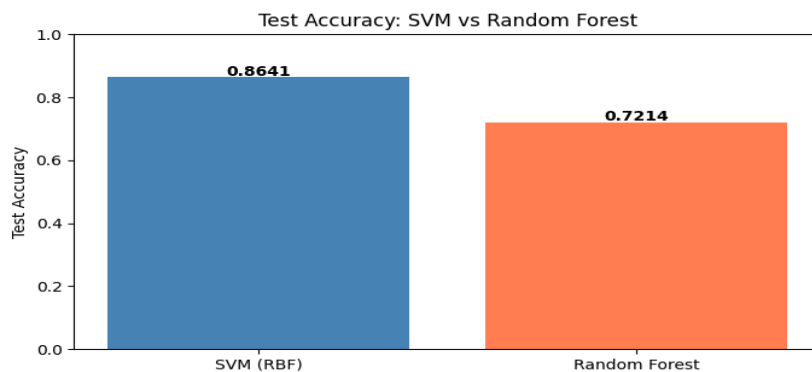


Figure 31: Test Accuracy: SVM vs Random Forest

2. Weighted F1-Score

This research evaluated the equilibrium between precision and recall throughout all categories by utilizing the Weighted F1-Score along with accuracy measurements. The results of the tests revealed

that the Random Forest model achieved a score of 0.7181, whereas the SVM model attained a Weighted F1-Score of 0.8638. This information indicates that the SVM model exhibits greater stability across all classification groups and aligns with the accuracy scores that were calculated earlier. Because it takes class distribution into consideration and offers a more complete view than accuracy alone, the F1-Score statistic is essential. These findings show that the SVM model can strike a balance between minimising classification mistakes and accurately identifying classes.

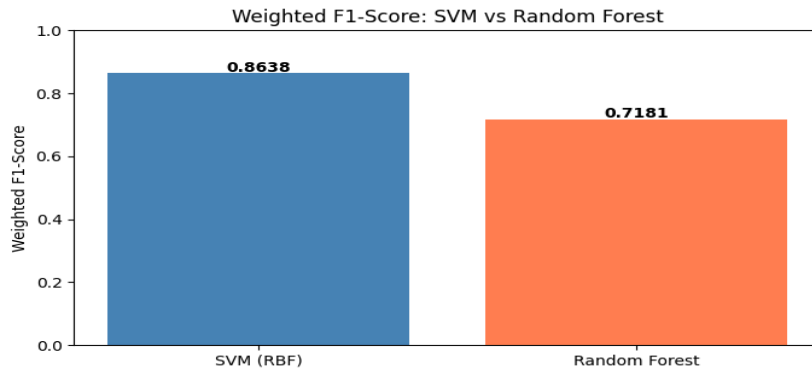


Figure 32: Weighted F1-Score: SVM vs Random Forest

3. Precision dan Recall

For every class, precision and recall measurements were used to provide a more thorough study. The VeryMildDemented class was the hardest to classify, with a recall of 0.78, while the ModerateDemented class performed the best in the SVM model, with a precision of 0.98 and a recall of 0.99. This implies that early Alzheimer's disease-related structural alterations in the brain are more similar to those of other classes, making them more difficult to differentiate. In the Random Forest model, both precision and recall metrics were generally lower than those of the SVM, especially for the VeryMildDemented class, which achieved a recall of only 0.54. These results indicate that the Random Forest model struggles to distinguish the early stages of Alzheimer's compared to the SVM model.

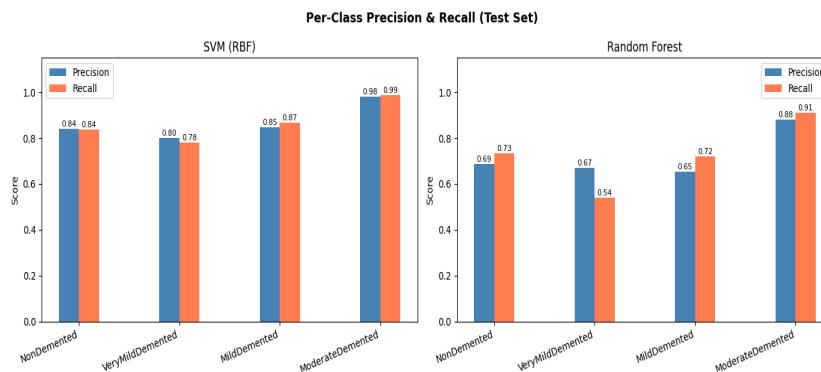


Figure 33: Per-Class Precision and Recall

4. Area Under the Receiver Operating Characteristic Curve (AUC-ROC)

The metric known as the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) was employed to evaluate the model's ability to distinguish between classes using a One-vs-Rest approach. The assessment indicated that the SVM model generated exceptionally high AUC scores for all categories: 0.9619 for NonDemented, 0.9479 for VeryMildDemented, 0.9775 for MildDemented, and 0.9996 for ModerateDemented. In contrast, the AUC values from the Random Forest model were lower for each category, measuring 0.8777, 0.8504, 0.8959, and 0.9833, respectively. AUC scores approaching 1 display that the model effectively distinguishes between positive and negative classes at various decision thresholds.

Alzheimer Classification on MRI Images

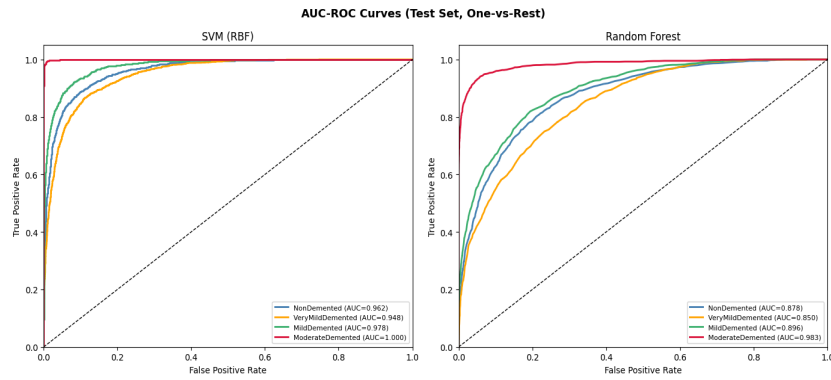


Figure 34: AUC-ROC Curves

5. Confusion Matrix

A confusion matrix was used to analyze the patterns of classification errors across the severity levels of Alzheimer's disease. The evaluation results showed that the SVM model was able to maintain a high recall rate across all classes, especially in the ModerateDemented class, with a recall of 98.87%. Most classification errors occur in clinically adjacent classes, such as between NonDemented and VeryMildDemented, as well as between VeryMildDemented and MildDemented. For example, 13.2% of the VeryMildDemented data was predicted as NonDemented and 8.3% was predicted as MildDemented. This pattern is understandable because anatomical changes in the early stages of Alzheimer's often still resemble normal conditions or later stages. Conversely, Random Forest showed a higher error rate, especially in the VeryMildDemented class, which had a recall of only 53.93%, resulting in many data points being misclassified into other classes.

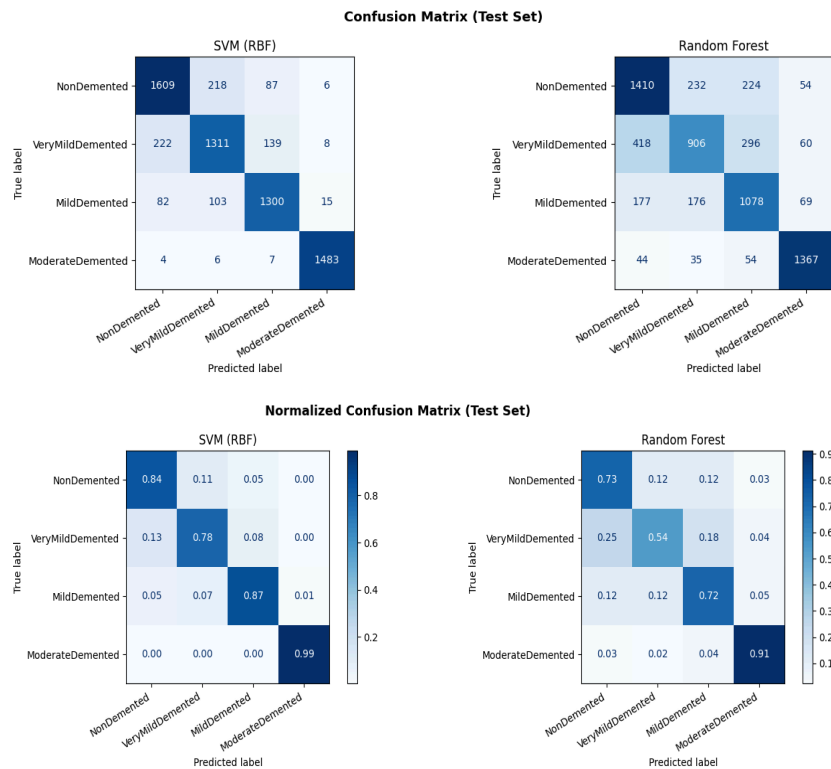


Figure 35: Confusion Matrix and Normalized Confusion Matrix

3.3 Stage 9: Interpretability

To determine how the retrieved features influenced the prediction process, an interpretability analysis was carried out in addition to assessing classification performance. To guarantee that the categorisation results were accurate, transparent, and clinically significant, Explainable Artificial Intelligence (XAI) methodologies were utilised. Principal Component Analysis (PCA), Random Forest Feature Importance, SHapley Additive exPlanations (SHAP), and t-distributed Stochastic Neighbour Embedding (t-SNE) comprised the analysis. These techniques were used to visualise the separability of Alzheimer's disease severity levels within the

feature space, select the most important features, and assess the contribution of each feature extraction approach. By combining performance evaluation with interpretability analysis, a more comprehensive understanding of the classification model was obtained.

1. SHapley Additive exPlanations (SHAP)

The contribution of individual features to the classification judgements produced by the Random Forest and Support Vector Machine (SVM) models was examined using SHapley Additive exPlanations (SHAP). By computing the average contribution of each feature to the final prediction, SHAP improves the transparency and interpretability of the model by providing a quantifiable measure of feature importance. Gabor-based features dominated the top-ranked feature importances, according to the Random Forest model's global SHAP analysis. Features with the greatest mean SHAP values included Gabor_317, Gabor_110, Gabor_138, and Gabor_13. A number of LBP characteristics also emerged as the most significant features, indicating that texture information obtained using local texture descriptors and multi-scale Gabor filters is essential for differentiating Alzheimer's disease severity levels. Furthermore, the per-class SHAP analysis revealed that different classes relied on different combinations of features, with ModerateDemented exhibiting the highest overall SHAP values, reflecting the more pronounced structural alterations associated with advanced stages of Alzheimer's disease.

The SHAP analysis revealed a marginally different pattern of feature relevance for the SVM model. GLCM_5 and GLCM_4 were the most significant features, followed by a number of LBP and Gabor features. This indicates that the SVM classifier benefited from complementary information recorded by Gabor and LBP descriptors while relying more heavily on second-order statistical texture characteristics. The top-ranked SHAP values contain features from all three extraction techniques, indicating that the suggested hybrid feature extraction framework effectively captures a variety of discriminative MRI texture characteristics. In general, the SHAP results verify that the combination of Gabor, LBP, and GLCM features aids in the classification process and offers significant insights into the texture patterns linked to varying degrees of Alzheimer's disease severity.

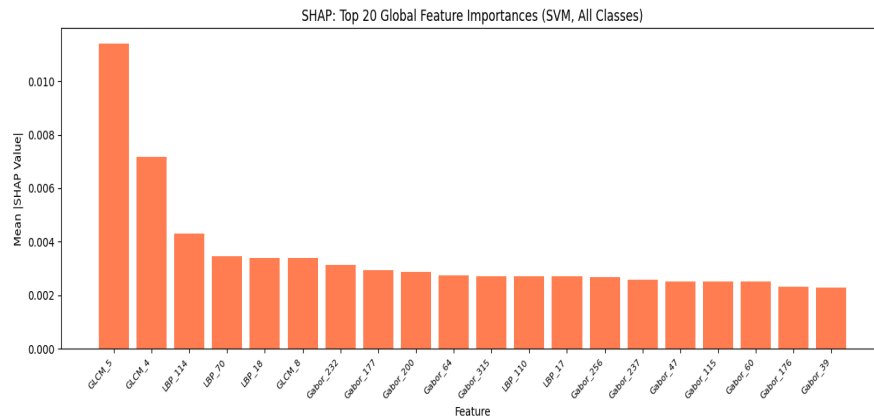


Figure 36: SHAP Global Feature Importance

2. RF Feature Importance

The most significant factors influencing the severity classification of Alzheimer's disease were found using Random Forest Feature Importance. The Gini Importance score, which is a measure of each feature's contribution to lowering impurity throughout the tree-splitting process, is used in this method. Both Gabor and LBP features dominated the top-ranked positions, according to the data, with LBP_28 emerging as the most significant feature (0.005800), followed by Gabor_13 (0.005649), Gabor_9 (0.005630), and LBP_54 (0.005588). Three of the top 10 features were from LBP descriptors and seven from Gabor descriptors, however none of the top ten features included a GLCM feature. These findings indicate that texture information extracted through multi-scale and multi-orientation Gabor filters, together with local texture patterns captured by LBP, provides the strongest discriminative power for differentiating Alzheimer's disease severity levels.

The feature significance scores were combined based on their extraction methods in order to assess each feature extraction method's contribution in more detail. Figure 37 illustrates that Gabor features accounted for 64.16% of the total relevance, followed by LBP features with 31.74% and GLCM features with just 4.10%. The predominance of Gabor features indicates that texture

characteristics based on frequency and orientation are very good at capturing structural changes linked to the advancement of Alzheimer's disease. Meanwhile, the substantial contribution of LBP features displays the importance of local texture variations in the classification process. In general, these findings verify that Gabor and LBP descriptors, which together offer the most pertinent data for Alzheimer's disease severity classification, are the main drivers of the suggested hybrid feature extraction architecture.

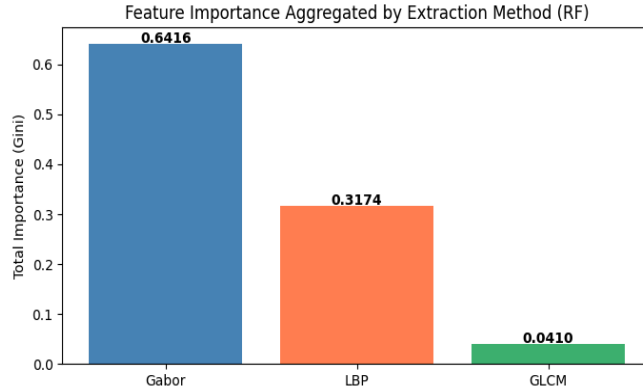


Figure 37: Aggregated Feature Importance by Extraction Method

3. Principal Component Analysis (PCA)

The distribution and separability of the retrieved feature space were visualised in a two-dimensional representation using Principal Component Analysis (PCA). PCA allows the identification of grouping patterns across Alzheimer's disease severity levels while maintaining the maximum amount of variance within the data by splitting the original high-dimensional feature space into two principle components. According to the resulting projection, the cumulative explained variance was 60.2%, with the first principal component (PC1) accounting for 42.4% of the total variance and the second principle component (PC2) for 17.8%. This shows that a significant amount of the data from the original hybrid feature space was preserved in the two-dimensional PCA representation.

The Ground Truth and SVM Prediction distributions displayed very similar spatial patterns, as seen in Figure 38, indicating that the SVM model was successful in capturing the feature space's underlying structure. Different clustering trends could still be seen even when there was some overlap between neighbouring classes, especially among the NonDemented, VeryMildDemented, and MildDemented data. The ModerateDemented class displayed a relatively more concentrated distribution, reflecting the more pronounced texture alterations associated with advanced Alzheimer's disease. The similarity between the ground-truth labels and the predicted labels further confirms that the proposed hybrid Gabor, LBP, and GLCM feature extraction framework provides a discriminative representation that enables the SVM classifier to effectively separate different severity levels of Alzheimer's disease.

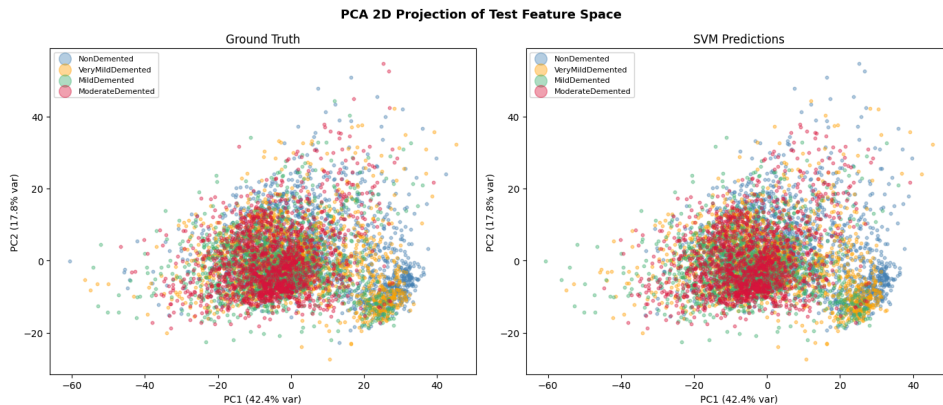


Figure 38: PCA 2D Projection of Test Feature Space for Ground Truth and SVM Predictions

4. t-distributed Stochastic Neighbor Embedding (t-SNE)

By maintaining local links between samples in a two-dimensional representation, the t-distributed Stochastic Neighbour Embedding (t-SNE) technique was used to better visualise the structure of the

retrieved feature space. Because t-SNE prioritises neighbourhood preservation over variance maximisation, it is more successful in identifying cluster patterns in complicated high-dimensional data. The hybrid feature set created from Gabor, LBP, and GLCM descriptors was used for the visualisation in order to evaluate how effectively the SVM model represented the underlying data distribution and investigate the separability of Alzheimer's disease severity levels.

The SVM model effectively learnt the underlying structure of the feature space, as seen in Figure 39, where the t-SNE projections of the Ground Truth labels and SVM Predictions display very comparable cluster distributions. There are a number of localised clusters visible, especially in the projection's right-hand section where samples are more closely clustered and exhibit a more distinct distinction from the rest of the data. However, there is still substantial overlap between the NonDemented, VeryMildDemented, and MildDemented groups, which reflects the similarity of textural features in nearby stages of the disease. The suggested hybrid feature extraction framework generates meaningful feature representations, allowing the SVM classifier to successfully maintain class relationships and differentiate Alzheimer's disease severity levels. This is displayed by the close correspondence between the ground-truth and predicted distributions.

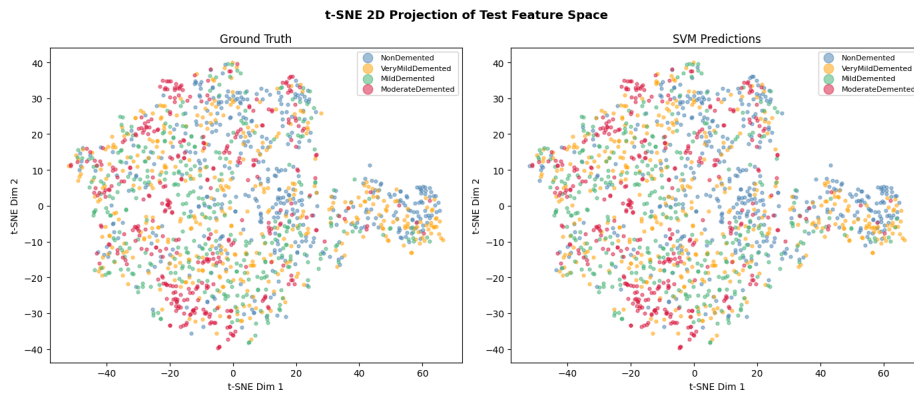


Figure 39: t-SNE 2D Projection of Test Feature Space for Ground Truth and SVM Predictions

4. Conclusion and Recommendation

4.1 Conclusion

Using a hybrid feature extraction method that includes Gabor, Local Binary Pattern (LBP), and Grey Level Co-occurrence Matrix (GLCM) descriptors, this study offered a framework for classifying the severity of Alzheimer's disease based on brain MRI images. The framework's extensive processing pipeline encompassed dataset validation, stratified data division, image preparation, data enhancement, feature identification, categorization, evaluation, and analysis of interpretability. The Support Vector Machine (SVM) utilizing an RBF kernel achieved superior performance on the test set, attaining an accuracy of 86.41% and a weighted F1-score of 0.8638, compared to the Random Forest classifier, which recorded an accuracy of 72.14% and a weighted F1-score of 0.7181. These findings suggest that the proposed hybrid feature representation effectively captures distinctive texture features linked to various phases of Alzheimer's disease.

The efficacy of the suggested framework was further validated by the assessment and interpretability analyses. The SVM model performed exceptionally well in detecting the ModerateDemented group, consistently achieving superior precision, recall, and AUC-ROC values across all classes. Gabor and LBP features made the biggest contributions to the classification process, according to SHAP and Random Forest Feature Importance studies, while PCA and t-SNE visualisations showed considerable class separability within the extracted feature space. The overall results display that the integration of Gabor, LBP, and GLCM features offers a dependable and comprehensible method for multi-class Alzheimer's disease severity classification from MRI images, despite some overlap remaining between clinically adjacent classes such as NonDemented, VeryMildDemented, and MildDemented.

4.2 Recommendation

The hybrid combination of Gabor, LBP, and GLCM features has shown good ability to differentiate Alzheimer's disease severity levels from MRI images, according to the data obtained. Nonetheless, classification mistakes between the NonDemented, VeryMildDemented, and MildDemented groups were still noted, indicating that the closeness of texture patterns makes early-stage Alzheimer's disease difficult to distinguish. In order to better capture small structural changes in the early stages of the disease, future research is advised to investigate alternate classification methods or new feature extraction techniques. Furthermore, a more thorough assessment of the suggested framework's robustness and generalisation capacity may be obtained by testing it on MRI datasets from various sources.

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